

view is provided by the accelerometer in the device (the same thing which auto-rotates the pictures and videos as you turn your phone). Depending on the application, either the server or the phone calculates what the camera is seeing the then superimposes the information with the placemarks.

HMDs, one of the prominent devices used for AR can facilitate quite a number of functions depending on what these are built for.



Infrared inputs built into AR systems for cars can help prevent accidents

HMDs which can be used by armed forces usually contain all components an AR browser for mobile would i.e. GPS, accelerometer and compass. The compass and accelerometer in the HMD can be used to track head movements to provide a more immersive experience. Another addition can be that of eye tracking. If the computer has the sense of where the user is looking, it

can improve the interface by adding more details to certain types of objects in the direction where the user is looking or help focus the user on only that area by causing other portions of screen to fade. However, in some cases the HMD does not have to contain a GPS such as when it is to be used in a closed environment and the position of the HMD is already known to other components of the AR setup. An engineer's workspace could be an example.

Additional input can be provided to the HMD using hand movements which can be done using special gloves which can be connected to the HMD using either wires or short range wireless methods (e.g. bluetooth). This enables gesture based input to the HMD for commanding the HMD without the use of voice. Gesture based input is used often for entertainment purposes wherein the user is allowed to play with virtual objects while hand tracking enables the user to interact with virtual objects. The day may not be far when you would be wearing a complete AR suit that adjusts what you see based on your blood pressure, skin humidity, muscle activity and other factors.

Infrared sensors can be used to aid night vision as well as to see objects normally hidden from eyesight using clever use of colors. These can be used to show the presence of an object which might be perilous and save a life.